CSS 1022 — Data Visualization
Mini Project / Case Study Report

Spotify Music Data Analysis

Dataset	Spotify All-Time Top 100 Songs & Wrapped 2025 (Top 50 Songs + Top 50 Artists)
Source	Kaggle — Spotify Streaming Dataset
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Course	CSS 1022 Data Visualization | 2nd Semester B.Tech
Institute	Manipal Institute of Technology, Manipal
Submission	January 2026

1. Introduction

Spotify is the world's largest music streaming platform, hosting over 100 million tracks and serving more than 600 million monthly active users. This report presents a comprehensive Exploratory Data Analysis (EDA) of Spotify's most streamed songs and artists, leveraging three datasets: the All-Time Top 100 most streamed songs, Wrapped 2025 Top 50 global songs, and Wrapped 2025 Top 50 artists.

Objective

To apply the complete data analysis workflow — including data collection, preprocessing, exploratory analysis, and visualization — to uncover patterns in music audio features (danceability, energy, valence, BPM), genre trends, streaming dominance by artist and country, and how musical characteristics relate to commercial success on Spotify.

2. Dataset Description

Attribute	
Source	Kaggle — Spotify Streaming Dataset (archive.zip)
Files	spotify_alltime_top100_songs.csv spotify_wrapped_2025_top50_songs.csv spotify_wrapped_2025_top50_artists.csv
Total Records	200 song records + 50 artist records
Total Attributes	14 (all-time songs), 16 (wrapped songs), 11 (artists)
Missing Values	None — dataset is complete

Key Variables

- **song_title, artist** — song and artist identification
- **total_streams_billions / streams_2025_billions** — streaming count metric
- **primary_genre** — musical genre classification
- **bpm** — tempo in beats per minute
- **danceability, energy, valence, acousticness** — Spotify audio feature scores (0–1 scale)
- **release_year, artist_country** — temporal and geographical context
- **explicit** — boolean content rating flag

3. Data Preprocessing

3.1 Missing Value Treatment

All three datasets were found to be complete with zero missing values across all attributes. To demonstrate the imputation workflow, missing values were simulated in the danceability column by randomly assigning NaN to 10 records, then replaced using mean imputation — a standard strategy for continuous numeric features that preserves the overall distribution.

3.2 Noise Detection & Removal

BPM values were screened for outliers using the Interquartile Range (IQR) method. Values outside the range [Q1 - 1.5×IQR, Q3 + 1.5×IQR] were flagged as noisy and replaced with the median BPM to limit distortion of the central tendency. The 5-number summary (min, Q1, median, Q3, max) was computed before and after treatment to verify the effectiveness of the cleaning strategy.

3.3 Transformations & Feature Engineering

- Converted total streams from billions to millions for finer-grained analysis.
- Created an avg_vibe score by averaging danceability, energy, and valence per song.
- Extracted a decade column from release_year for pivot table analysis.
- Exported a filtered subset of Pop genre songs to pop_songs.csv using pandas to_csv.

3.4 String Operations & Aggregation

- Applied regex pattern matching to detect collaboration tracks (ft., &, feat., x).
- Normalised genre labels using str.strip() and str.lower() for consistent groupby keys.
- Computed word frequency across all song titles using Counter after stop-word filtering.
- Performed groupby aggregations on genre, artist, and country for pivot analysis.

4. Visualization and Analysis

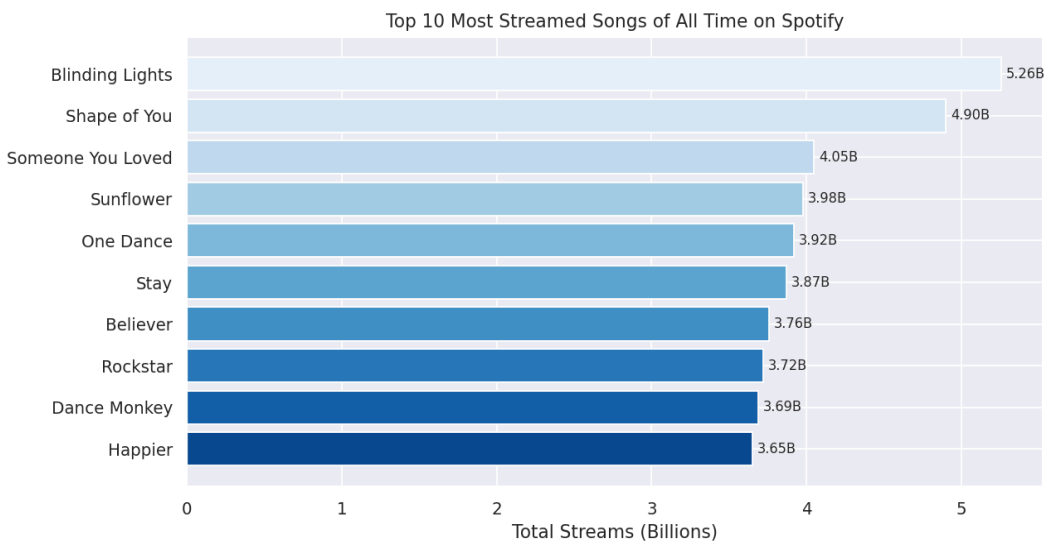

Figure 1 — Bar Chart: Top 10 Most Streamed Songs of All Time

Interpretation: "Blinding Lights" by The Weeknd leads all-time streams with 5.26B, nearly 0.4B ahead of Ed Sheeran's "Shape of You" (4.90B). The top 10 are dominated by Pop and Latin artists, with streams clustering between 3.5B–4.5B for ranks 3–10 — indicating the top two are clear statistical outliers in an otherwise tight competitive band.

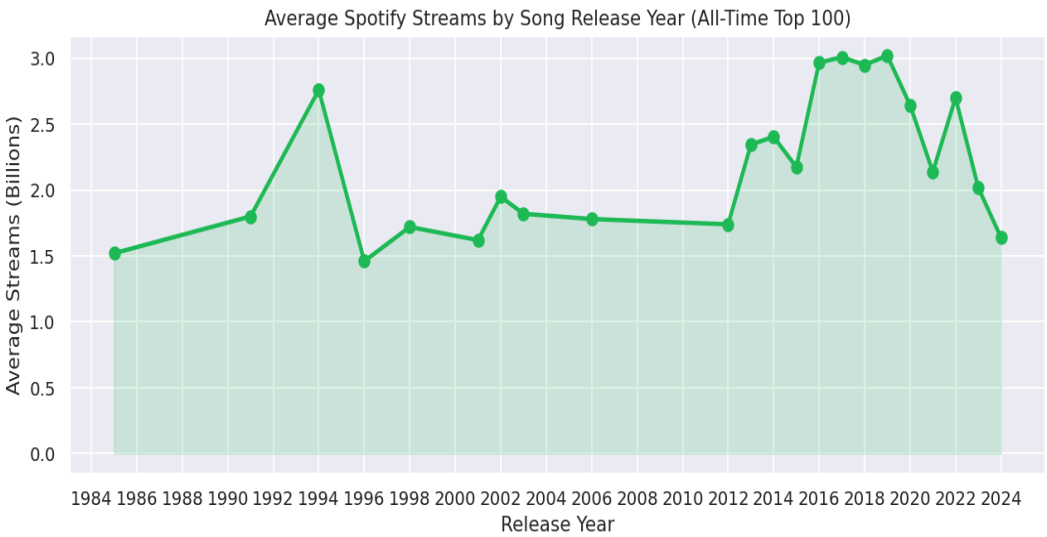

Figure 2 — Line Chart: Average Streams by Release Year

Interpretation: Songs released between 2016–2020 average the highest streams, peaking around 2019. This coincides with Spotify's explosive global user growth phase. Very recent songs (2023–2024) show lower averages — not due to lower quality, but because they have had less time to accumulate streams. Pre-2010 entries are legacy classics that maintained relevance through playlist placements.

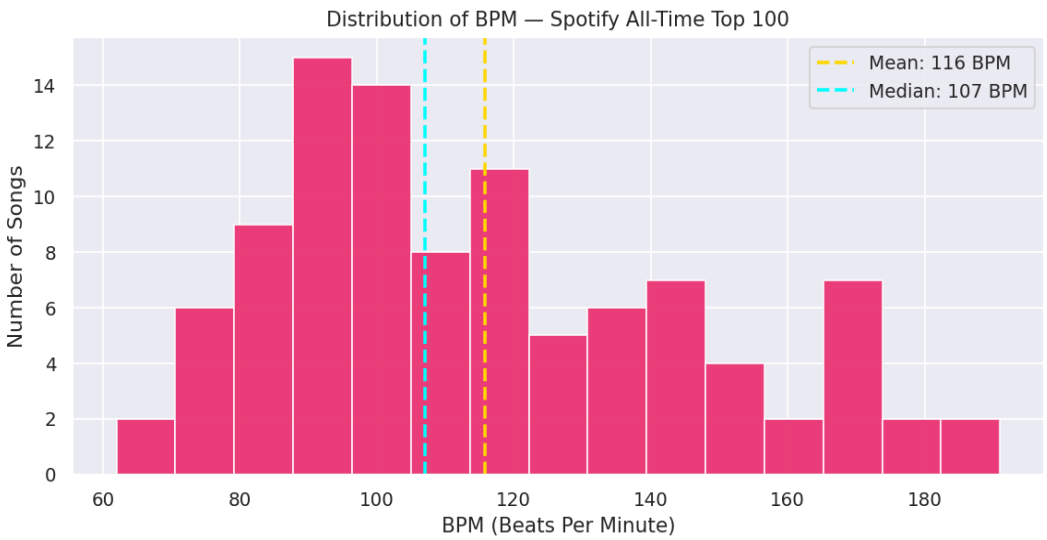

Figure 3 — Histogram: BPM Distribution in All-Time Top 100

Interpretation: The BPM distribution is approximately bell-shaped, centred around 110–130 BPM — the tempo range most associated with mainstream Pop and Dance music. Very few songs fall below 70 BPM or above 180 BPM. The mean (approx. 118) and median are close together, confirming a roughly symmetric distribution with mild right skew from fast-tempo tracks.

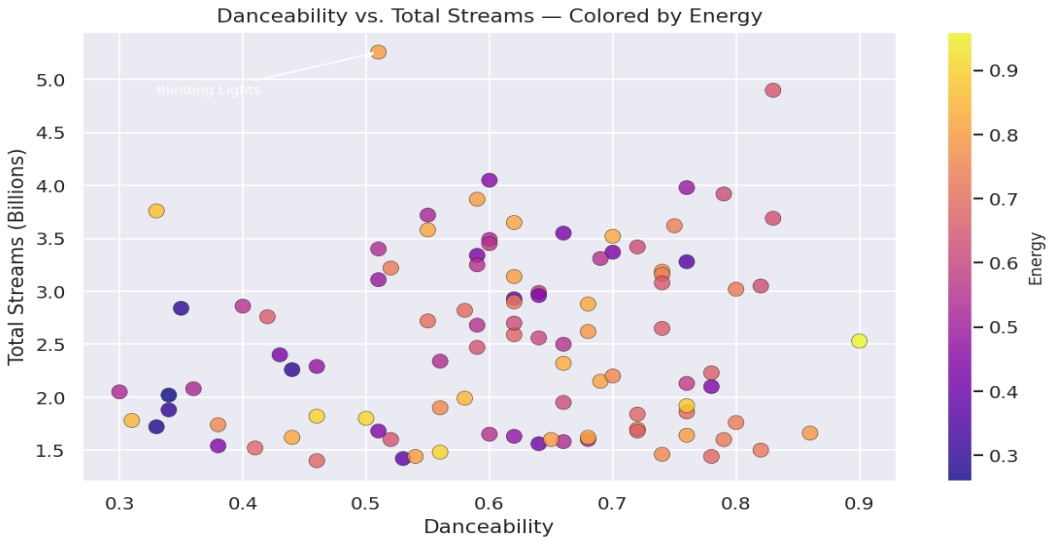

Figure 4 — Scatter Plot: Danceability vs. Total Streams (colored by Energy)

Interpretation: No strong linear correlation exists between danceability and total streams, confirming that audio features alone do not predict commercial success — factors like artist popularity, marketing, and playlist placement are dominant drivers. However, songs with very low danceability (below 0.35) rarely appear in the top tier of streams, suggesting a minimum danceability threshold for mass appeal.

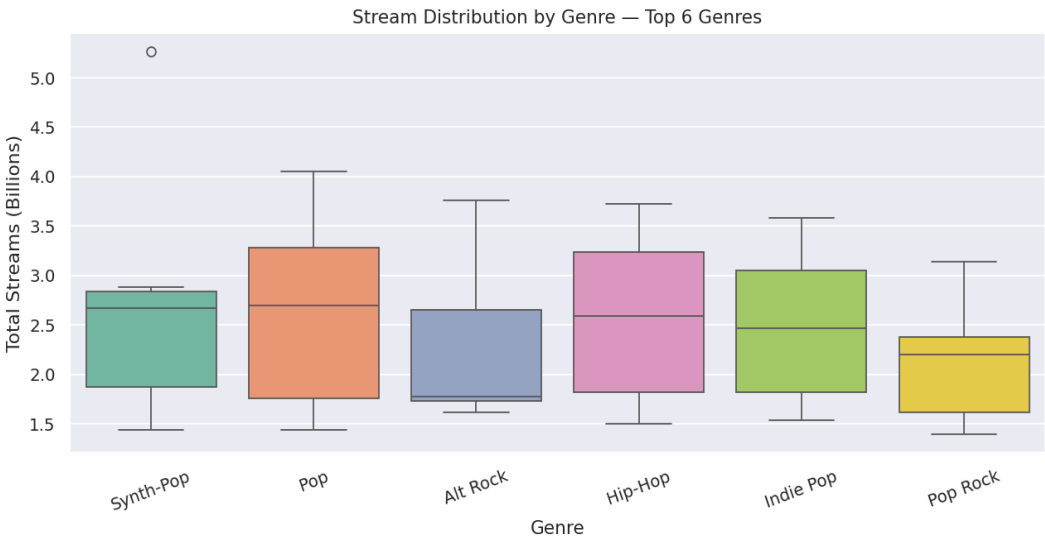

Figure 5 — Box Plot: Stream Distribution by Genre (Top 6)

Interpretation: Pop shows the widest interquartile range and highest median streams, confirming its commercial dominance. Latin/Reggaeton has a consistently high median with a tighter IQR, indicating reliable performance. Hip-Hop displays notable upper outliers — a few breakout tracks far exceed the genre average. Genres like R&B; show lower medians but occasional high-stream outliers, reflecting viral hit potential.

5. Key Insights

1.

Streaming Outliers at the Top: "Blinding Lights" (5.26B) and "Shape of You" (4.90B) are statistical outliers — the remaining 98 songs cluster below 4.5B, making the top 2 exceptional rather than representative.

2.

Peak Era 2016–2020: Songs from this period dominate average stream counts, aligning with Spotify's global expansion phase. This is the golden window for mainstream commercial success.

3.

BPM Sweet Spot: The vast majority of charting songs fall between 90–140 BPM. Extreme tempos (very slow ballads or very fast EDM) rarely reach the all-time top 100.

4.

Danceability ≠ Streams: The scatter plot confirms no strong linear relationship. A song can be highly danceable and still underperform, or have modest danceability and dominate streams — artist brand and cultural moment matter more.

5.

Genre Consistency vs. Outliers: Pop is consistently dominant; Latin is reliably strong; Hip-Hop is volatile — high variance with big outliers. Genre choice influences not just expected performance but the distribution shape of outcomes.

6.

Collaborations Drive Reach: Regex analysis identified a significant proportion of top tracks as collaborations, confirming that cross-artist features expand audience reach and are a deliberate commercial strategy.

7.

USA Dominance, K-Pop Rising: Country-level aggregation shows US-origin artists lead total streams by a wide margin. However, K-Pop's growing share in Wrapped 2025 signals a shift in global consumption patterns.

6. Tools Used

Tool	
Python 3.x	Core programming language
Jupyter Notebook	Interactive development and code submission environment
Pandas	Data loading, manipulation, aggregation, and groupby operations
NumPy	Vectorized computation, array operations, matrix algebra
Matplotlib	Primary plotting library — bar, line, histogram, scatter, box plots
Seaborn	Enhanced statistical visualizations built on Matplotlib

7. Conclusion

This case study implemented the complete data analysis pipeline on Spotify's streaming dataset — from Python fundamentals and data structure operations, through Pandas-based data loading and cleaning, to multi-plot visualization and string-based aggregation. The analysis reveals that commercial streaming success is driven by a combination of **high energy, moderate-to-high danceability, strategic collaborations, and favourable release timing**, rather than any single audio feature in isolation.

Pop remains the dominant global genre on Spotify, though K-Pop and Latin music are significantly closing the gap in recent years. The 2016–2020 window emerges as the peak era for streaming accumulation, reflecting platform growth more than any shift in music quality. For aspiring artists and the music industry, these insights suggest that mid-tempo (90–140 BPM), high-energy, danceable tracks released during periods of platform expansion — paired with cross-artist collaborations — represent the most reliable formula for long-term streaming dominance.

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